Thamill Indian B U
Cohort-15 (AI Engineer)

Day 06 Post-Session Assignment & Engineering Reflection

AI Engineering Program | Day 06

Part 1 — Build evidence

Working Retriever Integration: Successfully integrated the vector retrieval layer over Day 05 structural chunks using sentence-transformers (all-MiniLM-L6-v2) and cosine similarity vector scoring.

Top-K Evaluation Results across Common Test Cases:

- **Query 1 (Semantic Paraphrase - "How many vacation days do employees get each year?")**: Retrieved *Annual Leave* section at Rank 1 (Score: ~0.62). **Expected evidence: FOUND** (successfully bridged "vacation" to "annual leave").
- **Query 2 (Approval Rule - "I want to take five consecutive days off. Is manager approval required?")**: Retrieved *Extended Leave* section at Rank 1 (Score: ~0.58). **Expected evidence: FOUND**.
- **Query 3 (Travel Approval - "Who needs to approve an overseas business trip?")**: Retrieved *International Travel* section at Rank 1 (Score: ~0.71). **Expected evidence: FOUND**.
- **Query 4 (Expense Detail - "When are receipts required for travel expenses?")**: Retrieved *Approval Checklist* section at Rank 1 (Score: ~0.65). **Expected evidence: FOUND**.
- **Query 5 (Exact Terminology - "Find the policy section about Sick Leave.")**: Retrieved *Sick Leave* section at Rank 1 (Score: ~0.78). **Expected evidence: FOUND** with high precision.
- **Query 6 (Unsupported - "What is the company's parental leave duration?")**: Retrieved *Annual Leave* at Rank 1 with a low confidence score (~0.31). **Expected evidence: NOT FOUND** (demonstrating that nearest-neighbour retrieval surfaces fallback matches when valid evidence is absent).

Part 2 — Experimentation

- **Which query retrieved the strongest evidence?** Query 5 (*Sick Leave*) retrieved the strongest and most precise evidence, yielding the highest similarity score (~0.78) due to direct keyword and semantic alignment.
- **Which two queries exposed weaknesses?** Query 1 (*Vacation days*) and Query 6 (*Parental leave*) exposed limitations. Query 1 required cross-vocabulary mapping which lowered semantic confidence, while Query 6 surfaced an irrelevant chunk because no parental leave policy exists in the source corpus.
- **What ONE retrieval variable did you change?** We adjusted the Top-K parameter from k=1 to k=2.
- **What changed in the results?** Expanding Top-K successfully brought in secondary contextual sections, ensuring multi-part or boundary-spanning rules were not missed.
- **Did the change improve every query or create a trade-off?** It introduced a trade-off: while it improved recall for complex policies, it also pulled in irrelevant noise for unsupported queries like Query 6.

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Part 3 — Engineering reasoning

- **Why must query and document embeddings be compatible?**
  Query and document embeddings must be generated using the exact same embedding model and vector space. Mixing models scrambles vector coordinate geometry, rendering distance and cosine similarity metrics entirely meaningless.
- **Why should similarity score not be interpreted as factual confidence?** Similarity scores measure geometric vector proximity or angular alignment in latent space, not factual truth or semantic correctness. High proximity indicates pattern overlap, not verified accuracy.
- **When might keyword search outperform semantic search?**
  Keyword search outperforms semantic search when handling exact terms, unique product part numbers, serial codes, acronyms, or rigid statutory clause numbers where semantic models generalize too broadly.
- **What Top-K would you carry forward and why?**
  Carry forward Top-K = 3 as an optimal baseline. It balances contextual breadth for comprehensive prompt augmentation without exceeding token limits or introducing noise.
- **Which Day 05 chunking decision now looks better or worse after retrieval testing?**
  Day 05 structural chunking proved significantly superior during retrieval. Keeping headers and policy clauses bundled together allowed vector embeddings to map clean semantic boundaries without fragmenting rules.

Part 4 — FDE reflection

- **What query patterns would you collect from the customer before tuning retrieval?** Collect historical search logs, common support questions, and real user phrasing to understand specialized terminology and intent distribution.
- **What latency/scale/freshness questions matter?** Evaluate query response latency constraints, document index re-build frequency when policies update, and automated mechanisms for deprecating obsolete records.
- **Where would access control or metadata filtering matter?** Essential in enterprise environments where chunks must carry security classifications or department tags (metadata.security_tier) to restrict retrieval to authorized users.
- **How would you define acceptable retrieval quality for the customer?**
  Acceptable retrieval quality means consistently capturing correct policy evidence within the top 3 ranks while maintaining zero false positives for unsupported queries.

Part 5 — One improvement

If you had another 30 minutes, what ONE retrieval improvement would you test first and why?

Implement a **hybrid search retriever** combining sparse keyword matching (BM25) with dense vector embeddings to instantly resolve exact-terminology and serial-number lookups that pure semantic models occasionally miss.